# Skill Activity 2 - Analytical business reporting with SQL

```
use shopdb;
-- 1. Which products generate the highest revenue?
select
	p.product_id,
	p.product_name,
	sum(oi.quantity * oi.unit_price) as total_revenue
from products p
join order_items oi
	on p.product_id=oi.product_id
join orders o
	on o.order_id=oi.order_id
where
	o.status <> "Cancelled"
group by p.product_id
order by total_revenue desc;
```

OUTPUT:

| product_id | product_name | total_revenue |
|---|---|---|
| 1 | Laptop | 120000.00 |
| 2 | Headphones | 6000.00 |
| 6 | Jacket | 3000.00 |
| 3 | SQL Book | 2400.00 |
| 5 | T-Shirt | 1400.00 |

```
-- 2. How many orders are placed each month, and what is the monthly order trend?
select
	year(order_date) as order_year,
	month(order_date) as order_month,
	count(order_date) as total_orders
from orders
group by
	order_year,
	order_month;
```

OUTPUT:

| order_year | order_month | total_orders |
|---|---|---|
| 2024 | 4 | 1 |
| 2024 | 5 | 1 |
| 2024 | 6 | 2 |

```
-- 3. Which category has the highest number of products?
select
	c.category_id,
	c.category_name,
	count(p.product_id) as number_of_products
from categories c
join products p
	on c.category_id=p.category_id
group by
	c.category_id,
	c.category_name
order by number_of_products desc
limit 1;
```

OUTPUT:

| | category_id | category_name | number_of_products |
|---|---|---|---|
| ▶ | 2 | Books | 3 |

```
-- 4. Which product categories have the highest average product price?
select
	c.category_id,
	c.category_name,
	round(avg(p.price),2) as average_product_price
from categories c
join products p
	on c.category_id=p.category_id
group by
	c.category_id,
	c.category_name
order by average_product_price desc;
```

OUTPUT:

| | category_id | category_name | average_product_price |
|---|---|---|---|
| ▶ | 1 | Electronics | 31000.00 |
| | 3 | Clothing | 1850.00 |
| | 2 | Books | 533.33 |

-- 5. What is the total number of orders and total revenue generated in each order status?

```
select
        o.status,
        count(distinct o.order_id) as total_orders,
        sum(oi.quantity * oi.unit_price) as total_revenue
from orders o
join order_items oi
        on o.order_id=oi.order_id
group by o.status
order by total_revenue desc;
```

OUTPUT:

| status | total_orders | total_revenue |
|---|---|---|
| delivered | 2 | 127000.00 |
| shipped | 1 | 3800.00 |
| placed | 1 | 2000.00 |